§10.2 Calculus with Parametric Curves

Homework：3,15,25,31,39,51,53,59

$x = f(t)$, $y = g(t)$

1. Tangents：

   i. $\dfrac{dy}{dx} = \dfrac{\frac{dy}{dt}}{\frac{dx}{dt}}$, $if\ \dfrac{dx}{dt} \neq 0$

   ii. $\dfrac{d^2y}{dx^2} = \dfrac{d}{dx}\left(\dfrac{dy}{dx}\right) = \dfrac{\frac{d}{dt}\left(\frac{dy}{dt}\right)}{\frac{dx}{dt}}$

2. Areas：

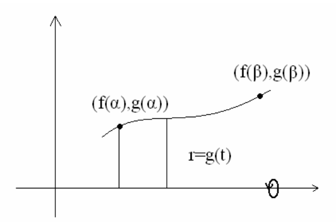
   $A = \int_a^b y\,dx = \int_\alpha^\beta g(t) f'(t)\,dt$

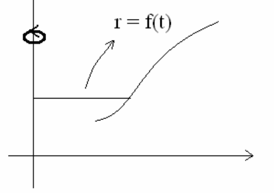
3. Arc Length：

   $L = \int_a^b ds = \int_a^b \sqrt{(dx)^2 + (dy)^2} = \int_\alpha^\beta \sqrt{\left(\dfrac{dx}{dt}\right)^2 + \left(\dfrac{dy}{dt}\right)^2}\,dt$

4. Surface Area：

$$S = \int_a^b 2\pi r\,ds$$
$$= \begin{cases} 2\pi \displaystyle\int_\alpha^\beta g(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\ dt \\ \\ 2\pi \displaystyle\int_\alpha^\beta f(t) \sqrt{\left(\frac{dx}{dt}\right)^2 + \left(\frac{dy}{dt}\right)^2}\ dt \end{cases}$$





Example 1： Given $x=t^2$, $y=t^3-3t$, $t\in R$. Find the equation(s) of the tangent(s) to the above curve at (3,0) .

Solution：

$$t^3-3t=0$$
$$\Rightarrow\ t=0\ (\text{不合})\ or\ \pm\sqrt{3}$$
$$\Rightarrow \frac{dy}{dx}=\frac{3t^2-3}{2t}=\frac{6}{2\left(\pm\sqrt{3}\right)}=\pm\sqrt{3}$$
$$\Rightarrow\ \text{Two tangents: } y=\sqrt{3}(x-3) \text{ and } y=-\sqrt{3}(x-3).$$

Example 2： Find the tangent to the cycloid $x=r(\theta-\sin\theta)$, $y=r(1-\cos\theta)$ at $\theta=\dfrac{\pi}{3}$.

Solution：

$$\frac{dy}{dx}=\left.\frac{\sin\theta}{1-\cos\theta}\right|_{\theta=\frac{\pi}{3}}=\sqrt{3}$$
$$x=r\left(\frac{\pi}{3}-\frac{\sqrt{3}}{2}\right)\ ,\ \ y=\frac{r}{2}$$
$$\Rightarrow\ \text{The tangent： } \sqrt{3}x-y=r\left(\frac{\pi}{\sqrt{3}}-2\right).$$

Example 3： Let $x=r(\theta-\sin\theta)$, $y=r(1-\cos\theta)$. Find $\dfrac{d^2y}{dx^2}$.

Solution：

$$\frac{dy}{dx}=\frac{\sin\theta}{1-\cos\theta}$$
$$\frac{d^2y}{dx^2}=\frac{d}{dx}\left(\frac{dy}{dx}\right)$$
$$=\frac{\frac{d}{dt}\left(\frac{dy}{dx}\right)}{\frac{dx}{dt}}$$

$$= \frac{\cos\theta(1-\cos\theta)-\sin^2\theta}{r(1-\cos\theta)^3}$$

$$= \frac{\cos\theta - 1}{r(1-\cos\theta)^3}.$$

Example 4： Find the area and arc length under one arch of the cycloid

$$x = r(\theta - \sin\theta)\ ,\ y = r(1-\cos\theta). \quad (0 \le \theta \le 2\pi)$$

Solution：

$$A = \int_0^{2\pi} r^2 (1-\cos\theta)(1-\cos\theta)\, d\theta$$

$$= r^2 \int_0^{2\pi} 1 - 2\cos\theta + \cos^2\theta d\theta = 3\pi r^2$$

$$L = r\int_0^{2\pi} \sqrt{(1-\cos\theta)^2 + \sin^2\theta}\ d\theta$$

$$= r\int_0^{2\pi} \sqrt{2-2\cos\theta}\ d\theta$$

$$= 2r\int_0^{2\pi} \sin\frac{\theta}{2}\ d\theta$$

$$= 8r.$$

Example 5： Show that the surface area of a sphere of radius r is $4\pi r^2$.

Solution：

$$S = \int 2\pi r ds = 2\pi \int_0^{\pi} r^2 \sin\theta d\theta = 4\pi r^2.$$